

Daniel A. Paradiso

Contact Information

Department of Physics
Syracuse University
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Education

Syracuse University, Syracuse, New York USA

Physics PhD Program

- Attended: 2022 - Present
- Advisor: Eric Coughlin

Syracuse University, Syracuse, New York USA

B.S. with Distinction, Physics, *Magna Cum Laude*, 2022

Research Interests

Shocks – Supernovae – Astrophysical Fluids – Accretion Physics – Black Hole Dynamics

Awards

National Science Foundation Graduate Research Fellowship, 2024-2027

Syracuse University Pre-Dissertation Summer Fellowship, Syracuse University Graduate School, 2023

Social Justice and Community Building Award, Syracuse University Department of Physics, 2022

Academic Excellence Prize, Syracuse University Department of Physics, 2022

Publications

On the Origin of Mass Ejection in Failed Supernovae

Paradiso, D.A., Vallejo, Sarah, & Coughlin, E.R., 2026, ApJ, 1004, 120

Gotta Go Fast: A Generalization of the Escape Speed to Fluid-dynamical Explosions and Implications for Astrophysical Transients

Paradiso, D.A. and Coughlin, E.R., 2025, ApJ, 985, 173

Low-energy explosions in a gravitational field: Implications for sub-energetic supernovae and fast X-ray transients

Paradiso, D.A., Coughlin, E.R., Zrake, J., & Pasham, D.R., 2024, ApJ, 961, 158

The Peak of the Fallback Rate from Tidal Disruption Events: Dependence on Stellar Type

Bandopadhyay, A., Fancher, J., Athian, A., Indelicato, V., Kapalanga, S., Kumah, A., **Paradiso, D.A.**, Todd, M., Coughlin, E.R., & Nixon, C.J., 2024, ApJL, 961, L2

Presentations

GECO Conference Series: The Extragalactic Transient Universe, May 2026

Title: *Shock Formation and Evolution in Failed Supernovae* (Oral)

IAU Symposium: Future Landscape of Astrophysical Transients, May 2026

Title: *Shock Formation and Evolution in Failed Supernovae* (Poster)

High Energy Astrophysics Division Meeting of the American Astronomical Society, October 2025

Title: *From Dud to Thud: Shock Formation and Evolution in Failed Supernovae* (Oral)

Cornell and Syracuse University Astrophysics Conference, August 2025

Title: *Shock Formation and Evolution in Failed Supernovae* (Oral)

European Astronomical Society Annual Meeting, June 2025

Title: *A Generalization of the Escape Speed to Fluid-dynamical Explosions* (Poster)

246th Meeting of the American Astronomical Society, June 2025

Title: *From Dud to Thud: Shock Formation and Evolution in Failed Supernovae* (Oral)

245th Meeting of the American Astronomical Society, January 2025

Title: *Gotta Go Fast: A Generalization of the Escape Speed to Fluid-dynamical Explosions and Implications for Astrophysical Transients* (Poster)

Research Oral Examination, Syracuse University, September 18, 2024

Title: *Life After Death: X-ray Transients from Failed and Sub-Energetic Core-Collapse Supernovae* (Oral)

Anticipating the Rising Tide of Tidal Disruption Events: Theory and Observations, UCSB KITP, April, 2024

Title: *X-ray Flashes from Failed and Sub-energetic Core-Collapse Supernovae* (Poster)

243rd Meeting of the American Astronomical Society, January 2024

Title: *Low-energy explosions in a gravitational field: Sub-energetic supernovae and fast X-ray transients* (Poster)

Teaching and Outreach

Camillus Middle School Astronomy Day, Camillus, NY

April 11, 2026

Frontiers of Science Lecture Series, Syracuse University

Title: *A Brief History of Adventures in Astronomy*
2023 - Present

Astronomy on Tap Public Lecture, Providence, RI

Title: *A Star is Torn: How Supermassive Black Holes Feed on Stars*
February 10, 2026

Nottingham Retirement Home Astronomy Club Lecture, Nottingham, NY

Title: *Gas, Stars, and Black Holes*

March 3, 2025

Teaching Assistant, Syracuse University

PHY 212: Electricity and Magnetism, *recitation* (2022 - 2024)

Syracuse University Research in Physics Summer Program Mentor, Syracuse University

2022 - Present

Physics Outreach Group, Syracuse University

2021 - Present

Holden Observatory Tour Guide, Syracuse University

August 2021 - Present

Guest Speaker at High School Physics Class, Watkins Glen High School

October 6, 2023

Physics Undergraduate Learning Assistant, Syracuse University

PHY 212: Electricity and Magnetism (2021 - 2022)

Museum of Science & Technology Kids Day, Syracuse, NY

February 22, 2020

Service

Physics Graduate Student Organization President, Syracuse University Department of Physics

2023-2024

Physics Graduate Admissions Committee Member, Syracuse University Department of Physics

2023-2024

Society of Physics Students Vice President, Syracuse University Department of Physics

2021-2022